

Term Paper
CSE-405, Computer Interfacing (Spring 2025)
Marks: 15

1. A CPU supports 5 interrupt lines: INTR1, INTR2, INTR3, INTR4, INTR5. They are connected in a daisy-chain priority scheme (INTR1 highest). If INTR3 is currently being serviced, and INTR1 becomes active, what will the CPU do if nested interrupts are enabled? What if they are disabled?
2. Suppose you are designing a digital audio playback system. The audio samples are stored in memory at 4000H–47FFH (2 KB), and the sound card's playback buffer is mapped at 5000H–57FFH. You want to use the Intel 8237 DMA controller, Channel 2 as the source and Channel 3 as the destination, to automate the transfer of these audio samples for playback. Write the procedure to program the 8237 DMA controller registers to perform this transfer in single-transfer mode.
3. A packaging factory uses a conveyor system driven by a stepper motor for precise movement of boxes. To increase resolution and reduce jerks in motion, the motor is operated in half-step mode. The stepper motor is interfaced with an 8086 microprocessor through the Intel 82C55 PPI. The motor windings are connected through driver circuits to the 82C55.
 - i) Explain the principle of half-step operation of a stepper motor with the excitation sequence required for clockwise rotation. Also, Draw the interfacing diagram of the stepper motor with the 8086 using the 82C55 PPI.
 - ii) Write an assembly language procedure to rotate the stepper motor one full revolution in half-step mode. Assume an 8-step sequence is required and that a suitable delay subroutine is available.